

UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy

DEGREE OF B.Sc. & M.Sci. WITH HONOURS

SECOND-YEAR EXAMINATION

03 00953 / 03 20266

LI ELECTROMAGNETISM 2

SEMESTER 2 EXAMINATIONS 2021/22

Time Allowed: 1 hour 30 minutes

Answer four questions from Section 1 and two questions from Section 2.

Section 1 consists of four questions and carries 40% of the marks for the examination.

Answer ***all four*** questions from this Section.

Section 2 consists of three questions and carries 60% of the marks.

Answer ***two*** questions from this Section. If you answer more than two questions, credit will only be given for the best two answers.

The approximate allocation of marks to each part of a question is shown in brackets [].

All symbols have their usual meanings.

Calculators may be used in this examination but must not be used to store text. Calculators with the ability to store text should have their memories deleted prior to the start of the examination.

A formula sheet and a table of physical constants and units that may be required will be found at the end of this question paper.

SECTION 1

Answer **all four** questions from this Section.

1. Write down the mathematical expressions for the divergence theorem and the curl (Stokes') theorem. Explain the meanings of the symbols. Using the divergence theorem, derive the differential form of the Gauss' law from its integral form. [10]

2. Starting from Maxwell's equations, demonstrate that electric and magnetic fields in free space, in the absence of charges or currents, obey the classical wave equations

$$\nabla^2 \vec{E} = \varepsilon_0 \mu_0 \frac{\partial^2 \vec{E}}{\partial t^2}; \quad \nabla^2 \vec{B} = \varepsilon_0 \mu_0 \frac{\partial^2 \vec{B}}{\partial t^2}. \quad [10]$$

3. Consider a monochromatic unpolarised plane electromagnetic wave incident on an air-glass interface. Define the term "plane of incidence". Using a sketch, explain how it is possible to obtain a linearly polarised wave using reflection and refraction. Which property of the Fresnel coefficients is responsible for this effect? Under what condition would this effect be observed? [10]

4. Give the definition of magnetisation, including the SI unit of magnetisation. Write down the relation between the $\vec{B}$ and $\vec{H}$ fields and magnetisation. Using this relation and the laws of magnetostatics, sketch the lines of the $\vec{B}$ and $\vec{H}$ fields produced by a nearly uniformly magnetised iron rod. [10]

SECTION 2

Answer **two** questions from this Section. If you answer more than two questions, credit will only be given for the best two answers.

5. A point-like electric dipole with a dipole moment p is located at a distance L from an infinite conducting plane. Consider two cases: (1) the dipole moment is parallel to the plane; (2) the dipole moment is perpendicular to the plane.

- (a) What is the electrostatic image of the dipole in each case? [6]
- (b) Find an expression for the force acting on the dipole in case (1), in terms of p and L . [10]
- (c) Find an expression for the force acting on the dipole in case (2), in terms of p and L . [10]
- (d) In each case, is the dipole attracted or repelled by the plane? Justify your answers. [4]

ANY CALCULATOR

6. Consider a spherical capacitor with inner and outer radii R_1 and R_2 , respectively, such that $R_2/R_1 = 1.25$. The capacitor is filled with a dielectric material, with relative permittivity depending on the distance to the centre of symmetry of the capacitor,

$$\varepsilon(r) = \alpha \left(\frac{R_1}{r} \right)^2, \quad \text{where } R_1 < r < R_2.$$

The potential difference between the inner and outer conductors is V_0 .

- (a) Show that the magnitude of the electric field between the conductors is $E = 4V_0/R_1$, and does not depend on r . **[8]**
- (b) Find the magnitude of the electric displacement field, $D = D(r)$, in terms of the constants α , R_1 and V_0 . **[8]**
- (c) Find the magnitude of polarisation, $P = P(r)$, in terms of the constants α , R_1 and V_0 . **[6]**
- (d) Find the volume density of induced polarisation charge, $\rho_p = \rho_p(r)$, in terms of the constants α , R_1 and V_0 . **[8]**

ANY CALCULATOR

7. (a) Explain the distinction between good and bad electrical conductors in the electromagnetic wave theory. Define a good conductor in terms of conductivity and frequency. Comment on the attenuation of electromagnetic waves in good and bad conductors. [10]

- (b) In a good conductor, electromagnetic wave propagation is governed by the equation

$$\nabla^2 \vec{E} - \sigma \mu_0 \frac{\partial \vec{E}}{\partial t} = 0,$$

where the symbols have their usual meanings.

- i. Obtain an expression for the wave number in the conductor for a linearly polarised electromagnetic wave. Hence show that the wave impedance is given by

$$Z = \sqrt{\frac{\omega \mu_0}{\sigma}} e^{i\pi/4}. \quad [12]$$

- ii. A wave propagating in vacuum is incident normally on the surface of a thick copper slab. Calculate the impedance Z for an incident wavelength $\lambda = 0.1$ mm. The conductivity of copper is $6 \times 10^7 \text{ S m}^{-1}$. [4]

- iii. Given that the wave impedance for free space is $Z_0 = 377 \Omega$, and that the reflection coefficient is $R = (Z - Z_0)/(Z + Z_0)$, determine the reflection coefficient and comment on your result. [4]

Formula Sheet

Vector calculus identities:

$$\nabla \times (\nabla \varphi) = 0; \quad \nabla(\nabla \times \vec{F}) = 0; \quad \nabla \times (\nabla \times \vec{F}) = \nabla(\nabla \cdot \vec{F}) - \nabla^2 \vec{F}.$$

Dipole electric field:

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0 r^3} \left[\frac{3(\vec{p} \cdot \vec{r}) \vec{r}}{r^2} - \vec{p} \right].$$

Dipole energy in uniform external field:

$$U = -\vec{p} \cdot \vec{E}; \quad U = -\vec{m} \cdot \vec{B}.$$

Cyclotron frequency and radius:

$$\omega_c = \frac{qB}{\gamma m}; \quad R = \frac{p}{qB}.$$

Polarisation charges:

$$\sigma_p = \vec{P} \cdot \vec{n}; \quad \rho_p = -\nabla \cdot \vec{P}.$$

Magnetisation currents:

$$\vec{j}_M = \vec{M} \times \vec{n}; \quad \vec{j}_M = \nabla \times \vec{M}.$$

Energy density of electric and magnetic fields:

$$w = \frac{1}{2}ED; \quad w = \frac{1}{2}BH.$$

The speed of light:

$$c = \frac{1}{\sqrt{\epsilon_0 \mu_0}}.$$

Plane wave properties:

$$\vec{B} = \frac{1}{\omega} (\vec{k} \times \vec{E}).$$

The Poynting vector:

$$\vec{N} = \vec{E} \times \vec{H}.$$

Skin depth:

$$\delta = \sqrt{\frac{2}{\mu_0 \sigma \omega}}.$$

Physical Constants and Units

Acceleration due to gravity	g	9.81 m s^{-2}
Gravitational constant	G	$6.674 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Ice point	T_{ice}	273.15 K
Avogadro constant	N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
[<i>N.B.</i> 1 mole $\equiv$ 1 <i>gram-molecule</i>]		
Gas constant	R	$8.314 \text{ J K}^{-1} \text{ mol}^{-1}$
Boltzmann constant	k, k_B	$1.381 \times 10^{-23} \text{ J K}^{-1} \equiv 8.62 \times 10^{-5} \text{ eV K}^{-1}$
Stefan constant	σ	$5.670 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$
Rydberg constant	R_∞	$1.097 \times 10^7 \text{ m}^{-1}$
	$R_\infty hc$	13.606 eV
Planck constant	h	$6.626 \times 10^{-34} \text{ J s} \equiv 4.136 \times 10^{-15} \text{ eV s}$
	$h/2\pi$	$\hbar$ $1.055 \times 10^{-34} \text{ J s} \equiv 6.582 \times 10^{-16} \text{ eV s}$
Speed of light <i>in vacuo</i>	c	$2.998 \times 10^8 \text{ m s}^{-1}$
	$\hbar c$	197.3 MeV fm
Charge of proton	e	$1.602 \times 10^{-19} \text{ C}$
Mass of electron	m_e	$9.109 \times 10^{-31} \text{ kg}$
Rest energy of electron		0.511 MeV
Mass of proton	m_p	$1.673 \times 10^{-27} \text{ kg}$
Rest energy of proton		938.3 MeV
One atomic mass unit	u	$1.66 \times 10^{-27} \text{ kg}$
Atomic mass unit energy equivalent		931.5 MeV
Electric constant	ϵ_0	$8.854 \times 10^{-12} \text{ F m}^{-1}$
Magnetic constant	μ_0	$4\pi \times 10^{-7} \text{ H m}^{-1}$
Bohr magneton	μ_B	$9.274 \times 10^{-24} \text{ A m}^2 (\text{J T}^{-1})$
Nuclear magneton	μ_N	$5.051 \times 10^{-27} \text{ A m}^2 (\text{J T}^{-1})$
Fine-structure constant	$\alpha = e^2/4\pi\epsilon_0\hbar c$	$7.297 \times 10^{-3} = 1/137.0$
Compton wavelength of electron	$\lambda_c = h/m_e c$	$2.426 \times 10^{-12} \text{ m}$
Bohr radius	a_0	$5.2918 \times 10^{-11} \text{ m}$
angstrom	$\text{\AA}$	10^{-10} m
barn	b	10^{-28} m^2
torr (mm Hg at 0 °C)	torr	$133.32 \text{ Pa (N m}^{-2}\text{)}$

Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches must be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to Student Conduct procedures.